Yuheng Wu

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Education

Shanghai University(China) **September 2019 – Present**

B.Eng in Cyber Security, Computer Science

GPA 3.81/4.0 (93.6/100) (rank 1st/27)

College Entrance Examination("Gaokao") **June 2019**

*Admitted as top **1.2%** ,Sichuan Province, P.R China*

among 650,000 students

Selected Course Projects

See more in my github repo.

Computer Security

- implement tunneling and encryption of a mini VPN, including TUN/TAP, IP tunneling, Routing, Authentication, and TLS/SSL
- TCP SYN flood attack, reset attack, session hijacking, Reverse Shell attack. ICMP redirect attack
- RSA encryption, Digital signature, analyze x.509 certificate
- using IPtables(netfilter) and LKM(loadable kernel module) to set up packets-filtering firewall

Deep Learning

- labs includes linear regression, classification, CNN, RNN, fine-tune transformer
- Reproduce the four basic components of *BasicVSR*. Implement bidirectional propagation, feature alignment, aggregation and upsampling(directly using conv layers and pixel shuffle)
- Using pre-trained SpyNet and EDVR-M to extract features following the setting of paper *BasicVSR*

Natural Language Processing

- From word2vec, seq2seq to RNN/Lstm based algorithm and finally to transformer
- implement Neural Machine Translation of Cherokee-to-English with RNNs
- implement transition based Neural Dependence Parsing
- Final project is to complete a question answering model based on *BiDaf*. Adding character-level embedding following paper *Convolutional Neural Networks for Sentence Classification*

Computer Networking

- implement a TCP protocol, core design(congestion control and fast recovery) is similar to TCP Reno
- retransmission strategy is following RFC6298

Deep Reinforcement Learning and Game Theory

- Implement and analyze Policy Gradient, Actor-Critic with General Advantage Estimate, Double Q-learning in mujoco
- Final project constructed a RL-powered self-driving car in the *Carla Simulator*. In this project, in order to make the system runs near real time on the limited computation resource, it abandon BEV and it only adopts a naive but tricky perception strategy–it leverages only one segmentation camera to cupture current environment and combine the Resnet extracted feature with the Carla's collision event detector's feature. In the decision stage, it generated actions via *SAC*.

Research Experience

THESBOT **Feb 2020 - Aug 2020**

Research intern at Research Institute of USV & Shanghai Key Laboratory of Intelligent Manufacturing and Robotics

- an industrial design for Hibot THESBOT-Dual: a semi-autonomous pneumatic robot for pipe inspection
- Redesign and optimize the structure of the robot, test the suitable materials
- Implement and test the maneuver strategy based on STM32.

RL-Adaptive Joint Task Configuration for Multi-Server Offloading **Feb 2022 - Dec 2022**

Research intern at Shanghai University GNSS Group

- this work is in preparation for *Globecom23*
- Inspired by Pensieve and Joint configuration Adaptation. Incorporate RL algorithm into improve network resource allocation in multi-servers edge computing offloading.
- Build a simulation environment to generate dynamic network to test different applications(pedestrian detection, video quality). The corresponding evaluation metrics are MR^{-2} and user QoE.
- incorporate super resolution backbone(EDVR) on the server side to enhance video quality.

Social & Intern Experience

Volunteer for National Cyberspace Security Week **Oct 2021**

Awarded as an outstanding volunteer by the committee.

Security Engineer Summer Intern [Qi An Xin Technology Group Inc] **August 2022**

top 2 information security company in China. Learning Security Response and information security for web, cloud and data centers. Graduated with honors.

Rewards & Recognition

Academic Merit Scholarship **Oct 2019 – Oct 2020**

- exceptional award for academic throughout the year(top3%).(Dean List)

Academic Merit Scholarship **Oct 2020 – Oct 2021**

- exceptional award for academic throughout the year(top3%).(Dean List)

Exceptional Student of Institute of Information Engineering, Chinese Academy of Sciences ***Sept 2022***

- Awarded for the exceptional and comprehensive performance of the three years of undergraduate and was recommended to postgraduate by the *Institute of Information Engineering of Chinese Academy of Sciences(CAS)*.

Exceptional Student in ShanghaiTech University Summer Camp **Sept 2022**

- Awarded for the exceptional and comprehensive performance in the summer campus and was recommended to postgraduate by the School of Information Science and Technology *ShanghaiTech University*.

Technical Skills

Languages

- Chinese(native)
- English(advanced)(TOEIC 865)
- Japanese(intermediate, equivalent to JLPT N2)
- Korean(beginner)

Programming

C, C++(std11/14), Python(data science and deep learning), Golang, html/css/javascript/react, Matlab, Latex, Docker

Mathematics background

- single/multi variable calculus
- linear algebra
- matrix analysis
- applied probabilistic theory
- information theory
- number theory
- discrete mathematics

Service

Student Member of China Computer Federation(CCF) **2021-Present**

Publications

one paper is in proceedings of Globecom23